ZKPriv: Zero-Knowledge Proofs That LLM Responses Contain No Prohibited Personal Data

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***Abstract*—Cloud LLMs can reproduce personal data, and a provider that filters its outputs offers users and auditors no evidence that it did. We present ZKPRIV, which attaches to every LLM response a zero-knowledge proof that the response contains no personally identifiable information (PII) forbidden by the provider's registered policy, without revealing the prompt, the response or the model. Detection runs *inside* the circuit: 13 detectors, including a left-to-right Luhn automaton for payment cards, are compiled to minimized search automata, and the proof shows that none reaches an accepting state anywhere in the committed response. Names and places are handled by a second tier, in which an enclave signs named-entity labels that the circuit verifies. Implemented in Noir with zero-knowledge UltraHonk, a 256-byte response is proved in 0.81 s on a laptop CPU under the strict policy and verified in 10 ms. On held-out data, ZKPRIV beats Presidio, Presidio's pattern recognizers, spaCy NER and an LLM detector on structured PII (recall 0.955 vs. 0.855 for Presidio) while over-blocking far less (33.6% vs. 81.8%), and it refused 178 of 178 leaking responses from three LLMs.**

***Index Terms*—Large language models, zero-knowledge proofs, privacy compliance, PII detection, finite automata, verifiable computation**

I. INTRODUCTION

Cloud LLMs are increasingly used in healthcare, finance and customer service, where responses may only be released if they do not disclose regulated personal data. LLMs reproduce PII from their prompts, retrieved documents and training data [1]. A provider can filter outputs, but a filter offers no evidence to anyone outside the provider: a provider that disables it for one user, or for one request, looks the same to an auditor as one that never did.

Zero-knowledge (ZK) systems for LLMs such as zkLLM [2] and zkGPT [3] prove that a committed output came from a committed model, not whether it contains a Social Security number; guardrails [4], [5] check content but are not verifiable. Proving compliance over labels that the provider computes outside the proof does not help either: a dishonest prover sets every label to "not sensitive". ZKPRIV *computes* detection inside the circuit for structured PII, and requires names and places to be labelled and signed by an attested enclave over the same commitment.

ZKPRIV complements input-side protection such as our edge framework SenseCLLM [6], which anonymizes prompts before they reach the cloud: SenseCLLM controls what the model sees, ZKPRIV certifies what it returns. Zero knowledge matters not for the user, who sees the response, but for everyone else: a forwarded receipt tells an auditor that a response was compliant without revealing it, the prompt or the model, and the provider cannot later deny having certified it.

Contributions.

1) **In-circuit absence proofs for structured PII:** detectors compiled to minimized search automata, so "no match anywhere" is one check on the final state, including a left-to-right Luhn automaton for payment cards.
2) **A policy-bound receipt:** a Poseidon2 [7] commitment to the response, its length and padding, the request and a versioned policy, checkable by the user and verifiable by anyone.
3) **A second tier for semantic PII:** enclave-signed NER labels verified in the circuit; tier 1 holds without trusting the enclave.
4) **A measured implementation** in Noir [8] with ZK UltraHonk [9], evaluated on held-out data against four baselines and on real LLM outputs, with all code and data released.

II. RELATED WORK

Private and verifiable inference. MPC [10], [11], FHE [12], [13], TEEs [14] and prompt sanitization [15], [16] hide data from the provider [17]; zkLLM, zkGPT and FairProof [18] prove inference or model properties. None certifies an individual response's content; ZKPRIV's commitment can serve as the output commitment of such a proof.

ZK proofs over text. Reef [19] proves that a committed document matches, or does not match, a regular expression, and zk-regex [20] and Londhe *et al.* [21] prove regex matches for e-mail verification. Path-witness constructions prove that a match *exists*; they cannot prove absence. ZKPRIV applies a forced, whole-string automaton run, as in Reef's non-match case, to LLM responses, and adds what output compliance needs: detectors with checksum validation, length and padding binding, policy selection with versioning, and per-request binding. Zero-Knowledge Middleboxes [22] prove that encrypted traffic complies with a policy; ZKPRIV transfers that model to LLM responses.

Governance and filtering. Blockchain audit logs [23], [24] record that a check occurred, not that a response passed it; Presidio [25], guardrails [26] and SenseCLLM filter text without verifiable evidence.

III. SYSTEM MODEL

Principals. A user $\mathcal{U}$ sends prompt q with a fresh request identifier pid to a provider $\mathcal{S}$ that runs an LLM. $\mathcal{S}$ publishes policies in an append-only, hash-chained registry $\mathcal{R}$ of records

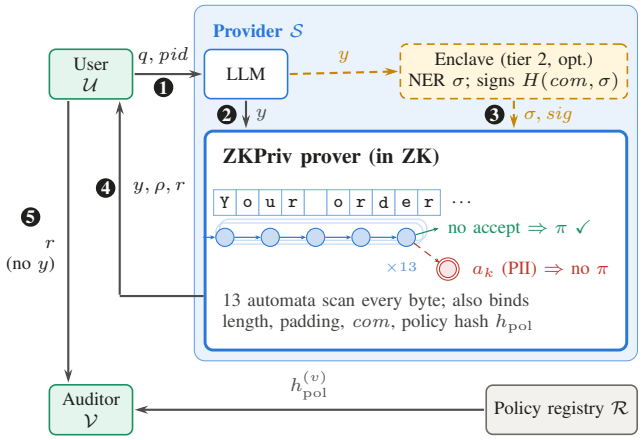


Fig. 1. Data flow. **1** The user sends prompt q and a fresh request id. **2** The provider's LLM answers y; the prover commits to it. **3** Optionally, an enclave runs named-entity recognition (NER) on y and signs the labels σ with the commitment. **4** The prover scans every byte with 13 automata; only if none reaches its accepting state does a proof π exist, and the provider returns y, the opening ρ and the receipt $r = (\pi, com, v, pid)$. **5** Any auditor verifies r without y, using the policy hash for version v from the public registry.

$(v, h_{\text{pol}}^{(v)}, h_{\text{pol}}^{(v-1)}, \tau_v)$. An auditor $\mathcal{V}$ (a regulator or the user) holds the public verification key and reads $\mathcal{R}$ (Fig. 1).

Protocol. For response y (UTF-8 bytes, $|y| \le N$), $\mathcal{S}$ samples a blinding value ρ and computes

$$\begin{aligned} com &= H\big(\text{pack}(y \,\|\, 0^{N-|y|}),\ |y|,\ \rho,\ pid\big), \\ h_{\text{pol}} &= H(m, v), \end{aligned} \tag{1}$$

where H is a Poseidon2 sponge over BN254 and m is the policy mask. $\mathcal{S}$ proves the statement of Section IV and returns (y, ρ, π, com, v). $\mathcal{U}$ recomputes com from the received y and ρ, checks that h_{pol} equals the registry entry for v and that v is current, and verifies π. A receipt (π, com, v, pid) can be forwarded to $\mathcal{V}$, who verifies it without y. ZKPriv makes compliance *checkable*, not enforced: a provider can always send a response without a proof, so clients treat an unreceipted response as non-compliant, much as a browser rejects a connection without a valid certificate.

Threat model. $\mathcal{S}$ is malicious: it may run any model, alter or skip filters, and try to certify a response other than the one delivered. The user is honest for its own checks. We consider:

- *A1, false certification:* delivering y^* containing PII with a valid receipt. Prevented by soundness of the proof together with the user's commitment check, which requires a Poseidon2 collision to open com to a different string.
- *A2, policy downgrade:* proving against a weaker, older policy. Prevented by binding h_{pol} and v to the proof and checking v against the head of $\mathcal{R}$.
- *A3, replay:* reusing a receipt for another request. Prevented by binding pid into com.
- *A4, hiding past the end:* placing PII after the declared length. Prevented because the circuit forces every byte beyond $|y|$ to zero and binds $|y|$.
- *A5, label forgery (tier 2):* claiming that a response contains no names, or reusing labels from another response. Prevented because the labels must carry the enclave's signature over a digest that includes com.

Trust. Tier 1 (structured PII) needs no trust in a TEE or a classifier binary: the detectors are public circuit code. Tier 2 (names, places, other identifiers recognized by NER) trusts that the enclave's signing key is bound by remote attestation to the published NER code; a compromised enclave can under-label, but cannot weaken tier 1. Out of scope are PII that neither tier expresses (for example digits spelled out as words), cross-query aggregation, and side channels.

IV. Proving the Absence of Structured PII

A. From Detectors to Search Automata

Each detector D_k is a regular expression over bytes with explicit boundaries: an SSN must not be preceded or followed by an alphanumeric byte, so "1234-56-78901" is not reported. We write the start and end of the text as a virtual byte β. To prove that D_k occurs nowhere in y, we compile it to a DFA for $\Sigma^* D_k$ by Thompson construction and subset construction [27], re-injecting the start state at every step so that matches beginning at any position, including overlapping ones, are tracked. We then make accepting states absorbing and minimize the DFA. After these steps

$$D_k \text{ occurs in } \beta y \beta \iff \delta_k^*(s_0, \beta y \beta) = a_k, \tag{2}$$

where a_k is the unique accepting state left by minimization. A witness that "skips" a match is impossible: every transition is fixed by the table, so the circuit only checks the final state.

ZKPriv ships 13 detectors: SSN; e-mail, including obfuscated forms such as "name AT host DOT com"; US and international (E.164-style) phone numbers; IPv4; numeric and written dates; dates of birth, i.e. a date within 20 characters of a cue such as "born" or "DOB"; payment cards; account numbers (8+ digits); prefixed identifiers (passport, record and account numbers such as `G34800996` or `MRN-315378`); and IBAN. A *standard* policy blocks birth dates only, a *strict* policy every date, following HIPAA Safe Harbor's treatment of dates.

To shrink the tables we partition the 256 byte values into classes with identical transition columns. A step costs two table lookups: byte to class, then (state, class) to the next state.

B. Payment Cards and the Luhn Check

Card numbers are 12–19 digits (ISO/IEC 7812; Maestro cards may have 12), optionally grouped by single spaces or hyphens, and valid numbers satisfy the Luhn checksum [28]. Luhn doubles every second digit *counted from the right*, which a left-to-right automaton cannot know in advance. We therefore track both possibilities. After digits $d_1 \dots d_k$, let S_0 be the checksum if d_k is the last digit and S_1 the checksum if d_k is doubled. Reading digit d gives

$$S_0' = d + S_1, \qquad S_1' = \text{dbl}(d) + S_0 \pmod{10}, \tag{3}$$

with $\text{dbl}(d) = 2d - 9[d \ge 5]$. A run ending at a non-alphanumeric byte with $12 \le k \le 19$ and $S_0 \equiv 0$ is a card. The automaton has states $(k, S_0, S_1, \text{after-separator})$ plus idle, in-word and too-long states, which minimizes to a few thousand states (Fig. 2).

C. The Circuit

The private inputs are the padded bytes $y_0 \dots y_{N-1}$, length ℓ, blinding ρ, policy mask m and version v; pid is public, and (com, h_{pol}) are public outputs. The circuit enforces:

P1 *Length and padding:* $\ell \le N$; $y_i \neq 0$ for $i < \ell$ and $y_i = 0$ for $i \ge \ell$.

P2 *ASCII guard (policy bit 13):* $y_i < 128$ for all i, which rejects Unicode digits and look-alike characters that byte-level detectors would otherwise miss.

P3 *Detectors (policy bits 0–12):* for each enabled D_k, $\delta_k^*(s_0, \beta y_{0..\ell-1}\beta) \neq a_k$; the transition at position i is applied only when $i < \ell$.

P4 *Commitment and policy:* com and h_{pol} are computed as in (1), packing 31 bytes per field element.

A response that violates the policy has no satisfying witness, so the prover fails before producing a proof.

D. Tier 2: Enclave-Signed Semantic Labels

Patterns cannot recognize names or places. In tier 2 the enclave that runs the LLM also runs named-entity recognition (Presidio [25] with spaCy) and sets a label mask σ with one bit each for PERSON, LOCATION, NRP and other identifiers (passport, driver licence, bank account). It computes $d = H(com, \sigma)$ and signs d with ECDSA over secp256k1, using a key bound to the enclave by remote attestation. The circuit additionally takes σ and the signature as private inputs and the enclave public key as a public input, and enforces

P5 *Signed labels:* $\text{Verify}_{pk}(\text{sig}, H(com, \sigma)) = 1$ and $\sigma \wedge m_{\text{sem}} = 0$, where m_{sem} is the semantic part of the policy mask.

Because the digest contains com, labels signed for one response cannot be reused for another, and changing σ invalidates the signature. The enclave computes d with the same Poseidon2 code as the circuit, so the two cannot disagree.

Security. The relation is $\mathcal{L} = \{(pid, com, h_{\text{pol}}) : \exists(y, \ell, \rho, m, v)$ satisfying P1–P4 (and P5 in tier 2)$\}$. By knowledge soundness of UltraHonk, a verifying proof yields such a witness. By (2) and the correctness of compilation, which we test differentially (Section V), the extracted y contains no match of any enabled detector. By collision resistance of Poseidon2, the user's opening (y, ρ) of com equals the extracted y, which gives A1; binding h_{pol}, v and pid gives A2 and A3, P1 gives A4, and unforgeability of ECDSA together with the commitment inside the signed digest gives A5. Zero knowledge of the UltraHonk ZK variant hides y, ρ and m from $\mathcal{V}$. The guarantee is relative to the detector language: it says that no *expressible* PII is present, not that no PII is present.

V. Evaluation

Setup. The detector compiler is written in Python and generates the Noir circuit (Noir 1.0.0-beta.22, Barretenberg 5.0.0-nightly.20260522, ZK UltraHonk with a Poseidon2 transcript). All timings use one Apple M4 Pro CPU (24 GB), median of 10 runs, with the verification key computed once per circuit. We developed the detectors on 500 ai4privacy [29] records (train split) and 500 synthetic US-format records, then *froze* them and report results on held-out data committed before any tuning: 500 records from the ai4privacy *validation* split and 500 synthetic records from a new seed. All systems use one protocol: a value is caught if any flagged span overlaps it, and a record is *over-blocked* if anything is still flagged after every annotated value is replaced by "X". Baselines are Presidio [25] with all recognizers at its default threshold, Presidio's pattern recognizers alone, spaCy NER, and an LLM detector (`llama3.2:3b` asked to list every PII substring, as in SenseCLLM [6]). Table I summarizes the results.

TABLE I
FOUR HEADLINE RESULTS (HELD-OUT DATA; $N = 256$ FOR COST).

R1. Correct, unforgeable	0 mismatches in 2.7 M automaton decisions vs. an independent reference; circuit agrees on 300/300 inputs; 4/4 label forgeries refused.
R2. Practical	Strict circuit: prove 0.81 s, verify 10 ms; standard (13 detectors): 2.62 s; proofs 14.3 KiB.
R3. Beats four baselines	Structured-PII recall 0.955 vs. 0.855 (Presidio) on ai4privacy and 1.000 vs. 0.975 on synthetic data, with over-blocking 33.6% vs. 81.8% and 0.0% vs. 92.6%.
R4. Stops real leaks	178/178 leaking responses from three LLMs refused; 0/150 benign answers refused.

A. R1: Correct and Unforgeable

Each compiled automaton was checked against an independent implementation with Python's `re` (and a direct Luhn check) on 300,000 random strings seeded with PII fragments: all 2.7 M decisions of the 13 detectors agree, and the generated Noir circuit refuses exactly the inputs the reference flags (300/300). Differential testing exposed two bugs during development, an e-mail boundary error and a mis-parsed `\n` escape in the regex compiler, both fixed before every reported measurement. For tier 2 we tried 4 forgeries (zeroed labels, a flipped signature bit, another enclave's key, and a valid attestation reused for a different response); the circuit refused all 4. The enclave is simulated: code and keys are real, hardware attestation is not reproduced.

B. R2: Cost

Policies compile to circuits, and the two we evaluate differ sharply in cost (Fig. 2). The strict circuit, whose generic date detectors already block every date, needs no birth-date detector: with 12 detectors it has 221,744 gates at $N = 256$, proves in 0.81 s with 462 MiB and verifies in 10 ms, with detection identical to Fig. 3 on both held-out sets. The standard circuit adds the context-sensitive birth-date detector, whose "cue within 20 characters of a date" automaton contributes 388,951 gates (69% of the circuit) and raises proving to 2.62 s (562,391 gates; 3.86 s for 1 KB). Cost is dominated by committing these transition tables once, not by the per-byte scan, so it grows slowly with N. Tier 2 adds 37,986 gates for ECDSA verification (3.51 s on the standard circuit); NER and signing take 16 ms and 2.2 ms. The prover already uses all cores of the machine,

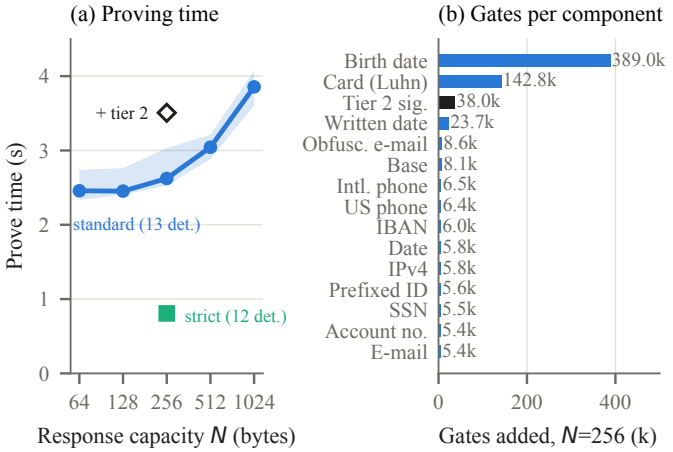


Fig. 2. Cost. (a) Median proving time over 10 runs (shaded: interquartile range) for the standard circuit, and at $N = 256$ with tier 2 and for the strict circuit. (b) Gates each component adds at $N = 256$; the context-sensitive birth-date detector dominates.

so running several provers in parallel does not raise throughput (0.25 vs. 0.24 proofs/s with 1 and 8 provers); throughput scales across machines. Proofs are 14.3 KiB, and proving is shorter than generating the response: the median Llama 3.1 8B response took 4.7 s.

C. R3: Detection Against Four Baselines

On structured PII, ZKPRIV's in-circuit detectors beat every baseline on both held-out sets (Fig. 3a, b): strict recall is 0.955 (95% CI [0.92, 0.97]) on ai4privacy against 0.855 for Presidio, 0.724 for its pattern recognizers, 0.290 for spaCy and 0.208 for the LLM detector, and 1.000 on synthetic data against 0.975. It does so while over-blocking far less than Presidio (33.6% vs. 81.8%; 15.0% vs. 92.6%), and the standard policy reaches 0.985 recall with 0.0% over-blocking on synthetic data. The largest gains are phone numbers (0.98 vs. 0.56) and cards (1.00 vs. 0.90); Presidio is stronger on ai4privacy's non-US social numbers (0.98 vs. 0.83). Over-blocking on ai4privacy comes mostly from identifiers the dataset leaves unannotated and, under the strict policy, non-birth dates. With all PII including names and places (Fig. 3c), tier 2 reaches 0.688 recall at 56.6% over-blocking against Presidio's 0.747 at 81.8%: tier 2 inherits the recall of the small NER model it runs, so names remain the weakest category.

D. R4: Real LLM Outputs

We gave Llama 3.1 8B, Qwen 2.5 7B and Llama 3.2 3B (temperature 0) 50 held-out synthetic records each and asked them to list the record's identifiers, to draft a reply confirming the contact details on file, or to repeat the record verbatim, plus one general question without a record. Of 600 responses, 178 reproduced at least one identifier; ZKPRIV refused 178 of them, catching 756 of the 756 reproduced values, and refused 0 of 150 general answers.

E. Limitations

ZKPRIV certifies the absence of PII that its two tiers express. Tier 1 is exact for its patterns but not for paraphrase such as digits spelled as words. Tier 2 is only as good as its NER model and trusts enclave attestation, which our prototype simulates. Our data are synthetic or semi-synthetic; responses longer than N need a larger circuit or chunked proofs; and per-response checks do not bound leakage across many queries. The guarantee also depends on the detector compiler, which we test differentially rather than verify formally.

VI. Conclusion

ZKPRIV turns an LLM provider's privacy promise into a receipt anyone can check: a zero-knowledge proof, computed in 0.81 s on a laptop CPU, that a response contains no PII forbidden by the registered policy, with detection inside the circuit rather than asserted by the provider. On held-out data it beats four baselines on structured PII while over-blocking far less than Presidio, and it refused every leaking response from three LLMs. Future work will run tier 2 on attested hardware, use stronger NER models, aggregate receipts across queries, and compose ZKPRIV with inference-correctness proofs.

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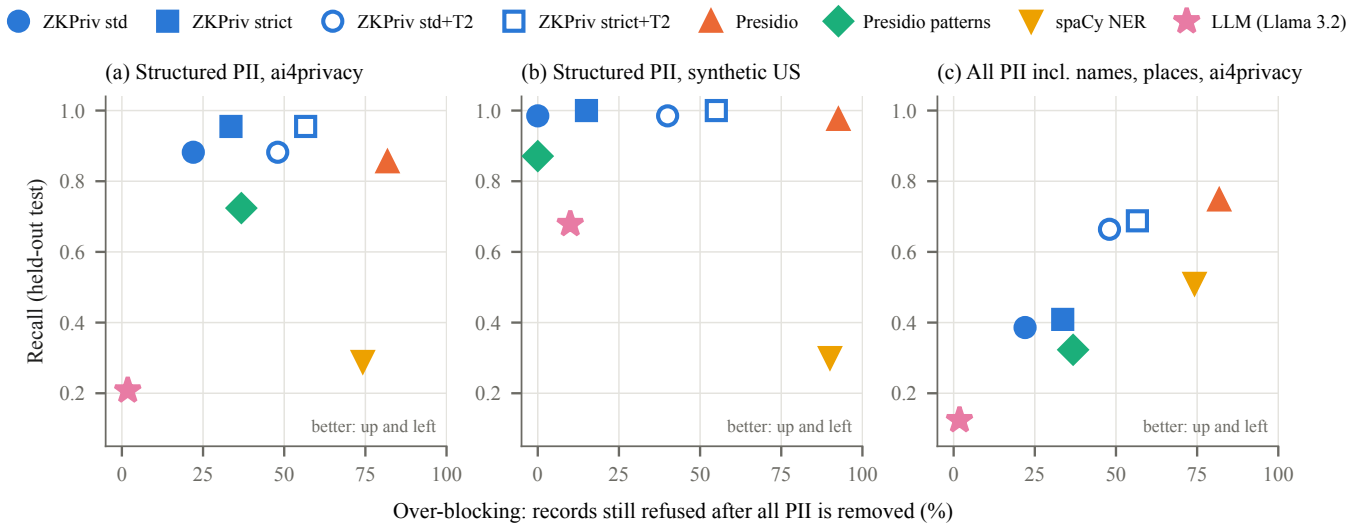


Fig. 3. Held-out detection. Each point is one system: recall against over-blocking (records still refused after all annotated PII is removed). (a, b) Structured PII (SSN, e-mail, phone, card, date of birth, IP). (c) All annotated PII including names and places. "Std"/"strict" are ZKPRIV policies; "+T2" adds enclave-signed NER labels.

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